

Beibei Liu

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Boston, MA - 02134, USA

EDUCATION

- **Ph.D. in Systems Engineering** Boston University
Advisor: Andrew P. Sabelhaus 2024 - 2029 (expected)
- **B.S. in Mathematics (Pure Mathematics Track)** The Chinese University of Hong Kong, Shenzhen
First Class Honors 2020 - 2024
 - GPA: 3.7/4.0

RESEARCH & ENGINEERING PROFILE

Robotics researcher with experience developing robotic systems from embedded hardware and software infrastructure to modeling, control, and experimental validation. Interested in building intelligent robotic systems that bridge research algorithms and real-world

RESEARCH EXPERIENCE

- **Soft Robotics Control Lab** [[Lab Link](#)] 2024 - present
Research Assistant Advisor: Andrew P. Sabelhaus
 - Designed, implemented, and experimentally validated a physics-based modeling and actuation framework for underactuated soft robots, *achieving over 60% improvement in trajectory tracking accuracy* over the baseline method in hardware experiments; resulting manuscript is accepted to IROS 2026.
 - Developed a multi-layer robotic software stack integrating ROS2, Python, C++ (PlatformIO), and control modules. Extended system functionality and experimental tooling to support controller development and hardware validation.
 - Designed and implemented the embedded control architecture for Arduino-based robotic hardware, improving system robustness and *establishing the laboratory's standard framework for future public release*.
 - Led the *fabrication and system integration* of a soft robotic prototype, including mechanical assembly, electrical integration, and experimental validation, enabling hardware testing for an ongoing laboratory publication.
- **Shenzhen Institute of Artificial Intelligence and Robotics for Society (AIRS)** [[Lab Link](#)] 2022 - 2024
Research Assistant Supervisor: Huihuan Qian
 - Developed learning-based disturbance prediction models using *wavelet neural networks and Gaussian Process Regression (GPR)* for marine robotic applications, achieving real-time prediction capability with *response times below 0.2 seconds* and supporting downstream motion-planning tasks.
 - Implemented and evaluated Model Predictive Control (MPC) algorithms for trajectory tracking of floating-base manipulators in MATLAB, conducting comparative studies with PID control to improve tracking performance and controller robustness.

SELECTED OPEN-SOURCE PROJECTS

- **ML-Based Soft Robot Surrogate Modeling and ROS2 Control Framework** [[GitHub Repository](#)]
 - Developed an end-to-end ROS2-enabled software framework integrating MATLAB simulation, PyTorch neural-network surrogate modeling, and Python communication for soft robot control experiments.
 - Designed and trained a lightweight multilayer perceptron (MLP) surrogate model to approximate nonlinear soft-robot dynamics for real-time control applications.
 - Integrated MATLAB and Python through asynchronous ROS2 communication to support automated experiment execution, stability monitoring, and real-time data exchange.
 - Designed a modular object-oriented software architecture with reusable control, communication, and visualization components to improve maintainability and extensibility.

ENGINEERING PROJECTS

- **Refrigerant Pump Design and Manufacturing Project** Fall 2025
Boston University ME559 Manufacturing Processes for Design and Production
 - Designed, manufactured, and validated a functional pump prototype using CNC machining, manual milling, waterjet cutting, and laser-cut fabrication processes.
 - Created a complete engineering workflow for a refrigerant pump system, including CAD design, material selection, detailed drawings, assembly procedures, and manufacturing planning.
 - Conducted tolerance stack-up analysis and developed supporting design documentation to ensure manufacturability and assembly reliability.
 - Communicated design decisions and tradeoffs through a formal engineering presentation on canned-motor pump architecture and manufacturing strategy.

PUBLICATIONS

J=JOURNAL, C=CONFERENCE, O=ONGOING, P=PRESENTATION AND POSTER

[C.1] B Liu, AK Dickson, R Jing, AP Sabelhaus (2026). **Trajectory Generation for Underactuated Soft Robot Manipulators using Discrete Elastic Rod Dynamics** . *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* 2026.

[J.1] R Xu, Z Jiang, B Liu, Y Wang, H Qian (2024). **Confidence-Aware Object Capture for a Manipulator Subject to Floating-Base Disturbances**. *IEEE Transactions on Robotics (T-RO)*, vol. 40, pp. 4396-4413, 2024, doi: 10.1109/TRO.2024.3463476.

[P.1] B Liu, R Jing, AK Dickson, AP Sabelhaus (2026). **Trajectory Generation for Underactuated Soft Robots using Discrete Elastic Rod Dynamics**. Poster presented at *the International Conference on Soft Robotics (RoboSoft)*.

[O.1] JP Garcia, AK Dickson, R Jing, B Liu, S Stokes, CV Hook, ML Anderson, AP Sabelhaus (Manuscript in preparation). **SRCBench: A Framework for Benchmarking Feedback Controllers in Soft Robotics**.

SKILLS

- **Programming Languages:** MATLAB, C++, Python, Java
- **Robotics:** ROS2, Simulink, AprilTag-based pose estimation, embedded systems (Arduino, PlatformIO)
- **ML:** PyTorch
- **Tools:** Git, Linux
- **Manufacturing:** 3D printing (Bambu Lab, Formlabs, nTop), CNC, CAD (Onshape)
- **Languages:** Mandarin Chinese (native), English (professional proficiency)

HONORS AND AWARDS

- **Teaching Assistant of the Year** April, 2026
Department of Mechanical Engineering, Boston University
 - One recipient selected annually across the department.
- **Academic Scholarship** 2020–2021
the Chinese University of Hong Kong, Shenzhen
 - Awarded to students ranked in the top 3% academically.
- **Dean’s List** 2020–2024
the Chinese University of Hong Kong, Shenzhen
 - Recognized for sustained academic excellence.

TEACHING EXPERIENCE

- **Teaching Assistant** Boston University
EK103: Computational Linear Algebra (Spring 2026)
 - Led weekly discussion sections and problem-solving sessions for undergraduate students.
 - Provided office-hour support, grading, and academic mentoring.
- **Teaching Assistant** The Chinese University of Hong Kong, Shenzhen
MAT2050: Mathematical Analysis (Fall 2023); MAT1012: Honors Calculus (Spring 2022)
 - Led weekly tutorial sessions and facilitated mathematical problem-solving discussions.
 - Provided academic support through office hours and student mentoring.